An official website of the United States government
Here's how you know

FULL TEXT LINKS

WILEY Full Text Article

FEBS Lett. 1990 Dec 10;276(1-2):75-7. doi: 10.1016/0014-5793(90)80510-p.

Finnish hereditary amyloidosis is caused by a single nucleotide substitution in the gelsolin gene

C P Maury [1], J Kere, R Tolvanen, A de la Chapelle

Affiliations

PMID: 2176164 DOI: 10.1016/0014-5793(90)80510-p

Abstract

The amyloid protein in Finnish hereditary amyloidosis is a fragment of the actin-filament binding region of a variant gelsolin molecule. Here we demonstrate, using polymerase chain reaction and allele-specific oligonucleotide hybridization analyses of genomic DNA, a single base mutation (G654---A654) in the gelsolin gene segment encoding the amyloid protein. The mutation is responsible for the expression of the variant (Asn187) gelsolin molecule in Finnish hereditary amyloidosis. The nucleotide substitution was found in all five unrelated patients with Finnish amyloidosis studied, but not in 45 unrelated control subjects. The mutation co-segregated with the disease phenotype in a family with Finnish amyloidosis. The results show that a single substitution in the gelsolin gene causes Finnish hereditary amyloidosis. The allele-specific oligonucleotide hybridization method provides a simple and accurate means of detecting this mutation.

PubMed Disclaimer

Related information

ClinVar
Gene (OMIM)
MedGen
MedGen (OMIM)
OMIM (cited)
PubChem Compound (MeSH Keyword)

LinkOut - more resources

Full Text Sources
Elsevier Science
Wiley

Other Literature Sources
The Lens - Patent Citations Database

Medical
MedlinePlus Health Information

Research Materials
NCI CPTC Antibody Characterization Program